An orbit is continuous falling

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Explain a circular orbit with gravity and sideways motion.

SUCCESS CRITERIA - I can...

1. Describe orbit as free fall rather than absent gravity.
2. Distinguish tangent velocity from inward force.
3. Connect circular speed to orbital radius.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “orbit” and “gravity” to explain your model or result. Sentence starters are optional.

orbit: a path around another body under gravity

gravity: an attractive interaction between masses

radius: distance from the center to the edge of a circle or sphere

01 Bring back an earlier idea

A ball launched horizontally at 6 m/s stays airborne 4 s. **Find** its landing point measured horizontally from the platform base.

Sentence starter: I found … [unit]. This value means … in this situation.

Marzano check - circle: ① ② ③

02 Explain a worked example

Reference: $GM/r^2 = v^2/r$; $v = \sqrt{(GM/r)}$.
Circular orbit outside a spherical body of mass M; r is orbital radius. Neglect atmosphere and other bodies.

Worked reasoning: A toy circular orbit has radius 4 m and speed 2 m/s. Its period is circumference/speed=$2\pi \times 4/2 = 4\pi$ s. Gravity continuously changes the velocity direction.

Your turn: Draw a quarter-orbit with tangent velocity arrows and inward gravitational arrows.

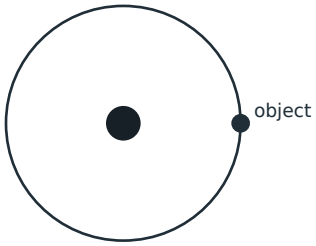
Sentence starter: The … in my model represents … . Its label shows … .

Marzano check - circle: ① ② ③

03 Look and explain

Draw the path of an orbiting object and its velocity/force arrows.
Explain how it can fall without immediately hitting the surface.

Sentence starter: My claim is … . The diagram/data show …, so … .



The center is the attracting body; the small dot on the circle is the orbiting object.

Marzano check - circle: ① ② ③

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Continue down each column.

Sentence starters are optional. You may plan in any language.

04 Build the solution

Model A has orbital period 8 s. How many orbits occur in 24 s? If an object covers a 40 m circumference in 8 s, find its average speed.

Sentence starter: I found ... [unit]. This value means ... in this situation.

Marzano check - circle: 1 2 3

05 Use the idea

Why does a rocket not need to keep firing forward to remain in an ideal circular orbit?

Word help: ideal = using stated simplifying conditions

Sentence starter: My claim is … . The diagram/data show …, so … .

Marzano check - circle: 1 2 3

06 Choose a method

A rover travels around a bend at a steady 6 m/s. **Describe** an observation that shows velocity changes even though the speed reading does not.

Sentence starter: I notice … . This supports … because … .

Marzano check - circle: 1 2 3

07 Exit check

Is an orbiting satellite accelerating if its speed is constant? **Draw** an arrow to support your answer.

Sentence starter: The … in my model represents … . Its label shows … .

Marzano check - circle: 1 2 3

08 STRETCH: Apply it another way

Compare two toy orbits with the same speed but radii 2 and 4 m. Which takes longer, and by what factor? **State** that this is a kinematic comparison, not necessarily two orbits around the same body.

Sentence starter: Both … . They differ in … because … .

Marzano check - circle: 1 2 3